PalmQ10

Installer Guide

Version 0.1.3-alpha

Private research release

Windows 10/11 with QGIS and Ubuntu on WSL2

Contents

1 About this release

PalmQ10 v0.1.3-alpha is a private research release for collaborating institutions — currently CEAM and the UPV Research Institute for Materials Technology.

This release adds HPC cluster execution. Earlier versions ran on a local workstation only. The heavy stages (WRF/WPS, WRF4PALM, PALM) can now be sent to a SLURM or SGE cluster, and connecting to one takes a single password per job rather than the repeated prompts of previous versions. It also carries several correctness fixes in the simulation chain — see *What's New* for the full list.

Building data currently covers Valencia and Lecco only

Building footprints and heights come from a prebuilt dataset covering **Valencia (Spain)** and **Lecco (Italy)** — the two areas currently provisioned for `eubds` access. The interface offers `eubds` as a building source, but it is an extract for those two cities, **not** a general European database. An area of interest elsewhere will still run, but **without realistic buildings**. Terrain, land cover, streets, and weather forcing are global; only buildings are restricted.

1.1 Other limitations worth knowing up front

- **Vertical stretching is not functional.** The control appears in the interface, but the feature is not yet implemented. Leave it set to `Off`.
- **Windows only** — Windows 10/11 with QGIS and Ubuntu on WSL2. There is no macOS or native Linux installer.
- **Lake physics is not enabled.** Water bodies use a fixed temperature.
- **Alpha status.** Some steps require user interaction, and some failures surface as raw model errors rather than friendly messages.

The *What's New* document lists these in full alongside everything that changed.

1.2 What you have been given

Item	What it is for
This guide	Everything needed to install and start using PalmQ10. **Follow it start to finish.**
Installer script	`install_palmq10_v0.1.3-alpha.ps1` — downloads everything else.
Uninstaller	`uninstall_palmq10_v0.1.3-alpha.ps1` — removes PalmQ10 if you ever need to.
User Guide	The interface, a worked two-hour Valencia example, and how to read the results. Start here once installed.
Cluster Connection Manual	Configuring and running on an HPC cluster. Only needed if you will use Rigel, Sirius, or another cluster.
What's New	Changes since the previous release.
Technical Guide	Architecture and internals. For developers; not needed to use PalmQ10.

Item	What it is for
Your access token	A GitHub token, sent to you separately. The installer asks for it. See Section 3.

1.3 What installation involves

1. Install three prerequisites yourself: QGIS, Ubuntu on WSL2, and 7-Zip.
2. Run the PalmQ10 installer script, which fetches everything else.
3. Complete first-use setup: access keys and enabling the plugin.

Budget around an hour, most of it unattended downloading. The release is roughly 6.7 GB.

Disk space

The installer requires at least **40 GB free** and recommends **50 GB**. Most of this is the geographic dataset, which expands to about 28 GB once extracted. Simulation outputs will need more on top of that.

2 Prerequisites

Install all three before running the installer. It checks for them and will stop and wait if any is missing.

2.1 QGIS

Install the QGIS **Long Term Release** for Windows from `https://qgis.org/download/`.

Open QGIS once after installing. This creates the user profile folder that the installer places the plugin into. If you skip this, the installer cannot find where to put the plugin and will pause.

2.2 Ubuntu on WSL2

PalmQ10 runs its simulation steps inside WSL. The tools it depends on — `rsync`, `ssh`, and `apptainer` — are Linux programs, so a Linux environment is required even though you drive everything from Windows.

Install Ubuntu from the Microsoft Store, then confirm it is running on WSL **version 2**, not version 1:

```
wsl -l -v
```

If the VERSION column shows `1`, open PowerShell **as Administrator** and run:

```
wsl --set-default-version 2
wsl --set-version Ubuntu 2
```

Verify again with `wsl -l -v`. Once it reports version 2, **open Ubuntu once and create your Linux username and password**. You will need that password during installation.

Your Ubuntu password is not your Windows password

During installation Ubuntu may ask for your `sudo` password. That is the Linux password you created inside Ubuntu, which is generally different from the one you use to log into Windows.

2.3 7-Zip

Install from `https://www.7-zip.org/`. The installer uses it to extract the geographic dataset, which ships as a split archive.

3 Getting your access token

PalmQ10 is distributed from a private repository, so downloading it requires a GitHub access token. The maintainers provide this to you directly — it is not something you generate yourself.

The token you receive grants read-only access to the release repository and nothing else. Treat it as a password:

- Do not commit it to a repository or paste it into a shared document.
- Do not forward it to colleagues. Request an additional token instead, so access can be revoked individually.
- Tokens expire. If installation fails with an access error, the token may have lapsed — ask for a replacement.

Keep the token somewhere you can find it

You will paste it into the installer when prompted. If you reinstall later, you will need it again.

4 Upgrading from an earlier version

Skip this section if this is a fresh installation.

You do not need to uninstall first, and **you do not need to move your work out of the way.** The installer replaces the previous version in place, keeping the two things that are yours:

- **Your projects.** `Documents\palmq10\jobs` — every simulation you have set up and every result you have produced — is set aside and restored afterwards.
- **Your access keys.** Your filled-in `ACCESS_KEYS_EDIT.txt` is kept, so you do not have to re-enter your credentials. It is *not* overwritten by the blank copy in the new release.

You will see both reported as the installer runs, first as "keeping" and then as "restored".

Everything else is deliberately replaced

The program code, the container images, and the geographic dataset are all reinstalled from scratch rather than reused. This is intentional — it guarantees no file from the previous version survives to cause confusing behaviour later. The practical cost is that the full 6.7 GB is downloaded again and the geographic data re-extracted, so allow the same time as a first installation.

4.1 What to expect

- **A full download again**, for the reason above. Budget the same time as a fresh install.
- **Your conda environment is updated, not rebuilt.** The existing `palmq10` environment is updated in place.
- **Your master password and cluster keys are unaffected.** These live outside the PalmQ10 folder, in your WSL home directory, and survive reinstallation.

Worth a backup anyway

Your projects are preserved automatically, but this is alpha software and results can represent a lot of compute time. If anything in `Documents\palmq10\jobs` would be painful to lose, copy it somewhere else before upgrading. It costs a minute and removes the question entirely.

4.2 If you would rather start completely clean

Run the uninstaller first — `uninstall_palmq10_v0.1.3-alpha.ps1`. It works on installations from any earlier version, since it targets the standard install locations rather than anything version-specific. Back up your projects and access keys first, exactly as above.

5 Running the installer

5.1 Download the installer script

From the shared folder or release page you were given, download:

```
install_palmq10_v0.1.3-alpha.ps1
```

Place it somewhere convenient, such as your Downloads folder. You only need this one file — it fetches everything else itself.

If you downloaded it from a shared drive

Windows marks files downloaded from the internet and may refuse to run them even after the execution-policy step below. If that happens, unblock the file first — in the same PowerShell window, before running the installer:

```
Unblock-File .\install_palmq10_v0.1.3-alpha.ps1
```

This affects only that one file.

5.2 Open PowerShell in that folder

In File Explorer, hold **Shift** and right-click inside the folder, then choose **Open PowerShell window here** (or **Open in Terminal**, depending on your Windows version).

5.3 Allow the script to run

Windows blocks unsigned scripts by default. Allow it for this session only:

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
```

This applies solely to the current PowerShell window and reverts when you close it. Your system-wide policy is unchanged.

5.4 Run it

```
.\install_palmq10_v0.1.3-alpha.ps1
```

If prompted about execution policy, type `O` to approve for the session.

The installer proceeds in this order:

1. Checks available disk space.

2. Checks for QGIS and Ubuntu on WSL2, pausing if either is missing.
3. **Asks for your GitHub token.** Paste it when prompted.
4. Downloads and installs the PalmQ10 code, QGIS plugin, and conda environment.
5. Installs Go and Apptainer inside Ubuntu. *Ubuntu may ask for your sudo password here.*
6. Downloads the container images and the geographic dataset.
7. Extracts the geographic data and verifies it.

Prerequisites are checked before the token is requested

This is deliberate — you are not asked for the token until the installer knows the rest of your system is ready.

5.5 What gets installed, and where

Component	Location
PalmQ10 program folder	`Documents\palmq10`
QGIS plugin	Your QGIS profile's plugin folder
Conda environment `palmq10`	Your Miniconda installation
Miniconda (if not present)	`%USERPROFILE%\miniconda3`
Container images	`palmq10\external\`
Geographic dataset	`palmq10\external\wps_geog\`
`PALMQ10_ROOT` variable	Set to your PalmQ10 folder

The installer avoids changing your global conda or PowerShell configuration. It creates a dedicated `palmq10` environment using temporary configuration files rather than modifying your existing setup.

Every download is checksum-verified

Each file's SHA-256 fingerprint is checked after download. A file interrupted by a dropped connection is detected and re-fetched rather than silently installed corrupted. Re-running the installer skips anything already present and verified, so an interrupted installation can simply be restarted.

6 First-use setup

6.1 Fill in your access keys

PalmQ10 downloads terrain, land cover, and weather data from public providers on your behalf. Several require a free account.

Open this file:

```
Documents\palmq10\documentation\ACCESS_KEYS_EDIT.txt
```

It ships blank. Fill in the credentials for the services you intend to use:

Provider	Needed for	Register at
Copernicus CDS	Weather forcing	`cds.climate.copernicus.eu`
Terrascope / ESA	Land cover	`sso.terrascope.be`
OpenTopography	ESA/COP30 terrain	`portal.opentopography.org`
Earthdata Login	NASA SRTM terrain	`urs.earthdata.nasa.gov`
AWS CLI	Canopy height	no account needed currently

The Copernicus CDS and Terrascope entries are needed for essentially every run. The two terrain providers are only required if you select those specific elevation sources — the default (FABDEM) needs no credentials at all.

Your credentials stay on your machine

Neither the PalmQ10 developers nor the developers of the underlying models receive them. If you are upgrading, copy your values across from your previous installation rather than registering again.

6.2 Enable the plugin in QGIS

Open QGIS and go to:

```
Plugins -> Manage and Install Plugins -> Installed
```

Tick the box for **PalmQ10 AOI**. The PalmQ10 icon appears in your toolbar.
If you had QGIS open during installation, restart it first.

6.3 Set a master password (cluster users only)

If you intend to run simulations on an HPC cluster, set your master password before your first cluster job. Skip this if you only run locally.
See the *Cluster Connection Manual* for the full procedure.

7 Verifying your installation

The quickest confidence check is a short simulation. The *User Guide* contains a worked two-hour Valencia example that exercises every stage of the pipeline while completing reasonably quickly. If that runs end to end and produces maps, your installation is sound.

Building data covers Valencia and Lecco only

Building footprints and heights come from a prebuilt dataset covering **Valencia (Spain)** and **Lecco (Italy)** — the two areas currently provisioned for `eubds` access. An area of interest elsewhere will run, but without realistic buildings. Terrain, land cover, streets, and weather forcing are global.

8 Troubleshooting

Problem	Resolution
Cannot access the GitHub release	The token is invalid, expired, or lacks access. Request a replacement from the maintainers.

Problem	Resolution
QGIS profile folder not found	QGIS has not been opened since installation. Open it once, then let the installer recheck.
Ubuntu not detected	Ubuntu is not installed, or not registered with WSL. Install from the Microsoft Store and open it once.
WSL1 detected	Convert to WSL2 using the commands in the prerequisites section. PalmQ10 will not run under WSL1.
Checksum mismatch	A download was corrupted, usually by an interrupted connection. Re-run the installer; it re-fetches only the affected file.
Insufficient disk space	Free space and re-run. Already-downloaded files are retained and skipped.
7-Zip not found	Install it, then let the installer recheck.
Script will not run, or Windows warns it came from another computer	The file is blocked because it was downloaded. Run `Unblock-File .\install_palmq10_v0.1.3-alpha.ps1`, then try again.
Sudo password rejected	Use your *Ubuntu* password, not your Windows password. If you have forgotten it, reset it from an Administrator PowerShell with `ubuntu config --default-user root`, then `passwd yourusername` inside Ubuntu.

8.1 Restarting a failed installation

The installer is safe to re-run. Files already downloaded and verified are skipped, so a run interrupted partway resumes rather than starting over.

9 Uninstalling

Download and run `uninstall_palmq10_v0.1.3-alpha.ps1` from the release page, using the same execution-policy step as for installation.

The uninstaller removes the PalmQ10 folder, the QGIS plugin, the conda environment, and the `PALMQ10_ROOT` variable. It does not remove QGIS, Ubuntu, 7-Zip, or Miniconda, since you may be using those for other work.

Save your work first

Project folders live inside the PalmQ10 directory. Copy anything you want to keep before uninstalling.

10 Further documentation

- *User Guide* — the interface, a worked example, and how to interpret results.
- *Cluster Connection Manual* — configuring and running on HPC.
- *What's New in v0.1.3-alpha* — changes since the previous release.

- *Technical Guide* — architecture and internals, for developers.

Supported systems. Windows 10 and 11, with QGIS and Ubuntu on WSL2. There is no macOS or native Linux installer in this release.

Licensing. Underlying components — notably PALM, WRF, GEO4PALM, and WRF4PALM — carry their own licence terms. Consult each project for conditions governing use and publication.